

cubst

Draw twisty puzzles in Typst

Version 0.1.0

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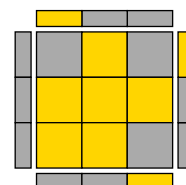
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1 Quick start

```
#import "@preview/cubst:0.1.0": *
```

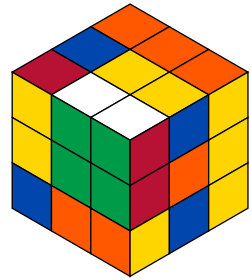
`case()` builds the state an algorithm solves; `draw()` draws it:

```
#draw(case("R U R' U R U2 R'"), view: "oll")
```

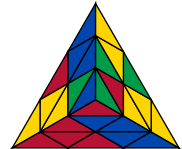


`cube()` builds a scrambled puzzle; `event:` picks the puzzle:

```
#draw(cube(scramble: "R U F2 D' L B'"), view: "full")
```



```
#draw(cube(event: "pyraminx",
  scramble: "U L R' B u"), view: "tip")
```



Everything follows this pattern: build a **state** (Section 2), then draw it (Section 3).

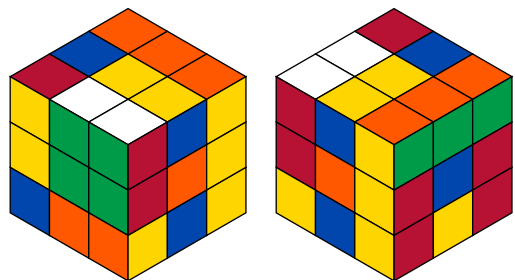
2 Building a state

```
cube(event: "3x3", scramble: none, inverted: false, scheme: auto, options: ({}))
case(alg, event: "3x3", scheme: auto, options: ({}))
solved(event: "3x3", scheme: auto, options: ({}))
apply(c, alg)
```

- event which puzzle; case-insensitive, "3x3x3", "3*3" and "3*3*3" are accepted for "3x3"
- scramble moves applied to the solved puzzle, in the order written
- inverted apply the inverse instead, giving the state that scramble **solves**
- scheme which color name sits on which face when solved; auto = the puzzle's default (Section 2.2)
- options settings of one puzzle, currently only the megaminx cut (Section 4.3); others are errors

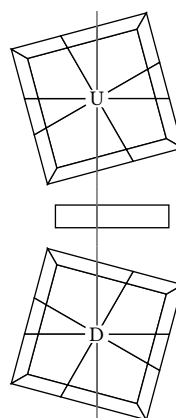
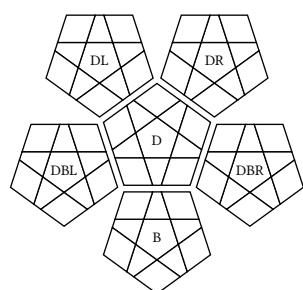
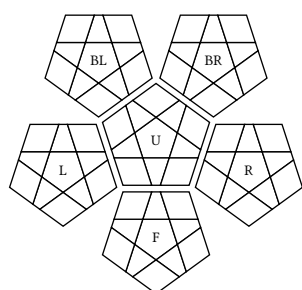
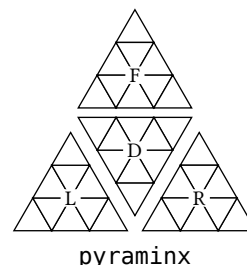
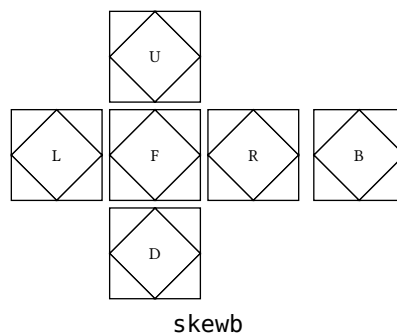
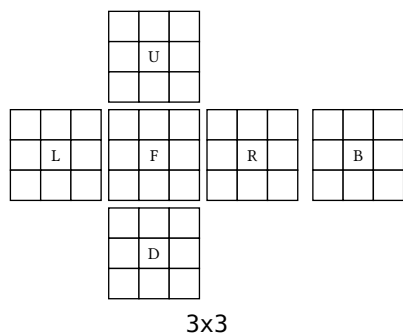
case(alg) is cube(scramble: alg, inverted: true): use it for every algorithm diagram (OLL, PLL, ...). apply adds moves to a state and returns a new one; states are plain values, never modified in place:

```
#let c = cube(scramble: "R U F2 D' L B'")
#draw(c, view: "full")
#draw(apply(c, "y"), view: "full")
```



event	puzzle	faces
"2x2", "3x3", ... any "NxN"	N×N cube	U D F B R L
"skewb"	skewb	U D F B R L
"pyraminx"	pyraminx	F L R D
"megaminx"	megaminx	U F R BR BL L D DR DBR B DBL DL
"square1"	Square-1	U D (the scheme still names four sides F B R L as cubes)

Where the faces are, as the net view unfolds them (labels: "faces" writes the names). On the Square-1 the piece between the layers is the front of the equator (the middle layer); its F B R L sit as on a cube:



2.1 Notation

WCA notation, plus whole-puzzle rotations where the WCA has none. Spaces optional where unambiguous (RUR'U'). (R U R' U')³ repeats a group, (. .)' inverts it. Commutators and conjugates (except for Square-1) nest freely: [A, B] is A B A' B'; [A: B], or A: B without brackets, is A B A'; so F: [R, U] and [F: [R, U]] both read F R U R' U' F'. Like (. .), a bracket can be repeated or inverted: [R, U]³, [R, U]'. Bad input fails with a cubst: message.

N×N cubes

R L U D F B

' 2 2'

Rw r 3Rw

M E S

x y z

skewb

R L U B

y

pyraminx

U L R B

u l r b

face turns, clockwise seen from that face

counter-clockwise, half turn

wide turns; leading number = layer count

slices, following L, D, F

whole-cube rotations about the R, U, F axes

120° clockwise about the DBR, DFL, UBL, DBL corners, seen from the corner; UFR never moves

whole-puzzle quarter turn, clockwise seen from above

120° clockwise about that vertex, seen from it; tip and layer tip only

pyraminx

y z whole-puzzle 120° turns: y about the U vertex, clockwise from above; z about the F centre, clockwise from the front

megaminx

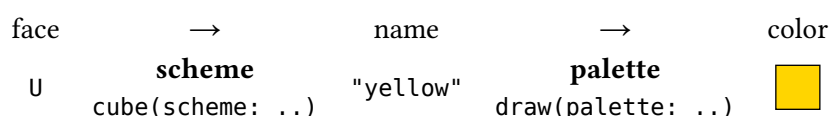
U F R L BL BR DR DL DBR DBL B D face turns, one fifth; suffixes ' 2 2 '
R++ R-- all but L, two fifths, clockwise / counter-clockwise seen from the right
D++ D-- all but U, two fifths, seen from below

Square-1

(x, y) top layer x·30°, bottom layer y·30°, clockwise
/ right half 180°; a compile error when a corner blocks it

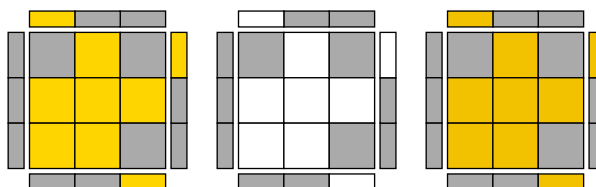
2.2 Color scheme

A sticker holds a color **name**, such as "yellow", not a color. Two lookups turn a face into ink:



The scheme is part of the state: it decides which stickers end up where. The palette (Section 3.3.1) is part of the picture: it only changes the hues. Below, the same algorithm with the default scheme, with white on top (white is now “the top color”), and with another palette (same state, different yellow):

```
#let white-up = (..default-scheme,  
  U: "white", D: "yellow", R: "red",  
  L: "orange")  
#draw(case(sune), view: "oll")  
#draw(case(sune, scheme: white-up),  
  view: "oll")  
#draw(case(sune), view: "oll",  
  palette: (..colors, yellow:  
    rgb("#f2c200")))
```



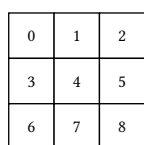
Every name in a scheme must exist in the palette. Views show faces by position, not by color; only the oll, f2l and obl masks read the scheme, to find the top (and bottom) color. The defaults:

N×N cubes, skewb	(U: "yellow", D: "white", F: "green", B: "blue", R: "orange", L: "red")
pyraminx	(F: "green", L: "red", R: "blue", D: "yellow")
megaminx	(U: "white", F: "green", R: "red", BR: "blue", BL: "yellow", L: "purple", D: "grey", B: "lime", DBL: "orange", DL: "lightblue", DR: "beige", DBR: "pink")
Square-1	(U: "yellow", D: "white", F: "red", B: "orange", R: "green", L: "blue")

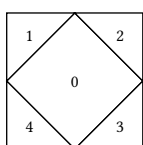
2.3 Reading a state

```
face(c, "U")           // stickers of a face: rows on a cube, else a flat array
sticker(c, "U", 0, 2)  // one color name, or none when hidden
is-solved(c)           // whether the cube is solved
```

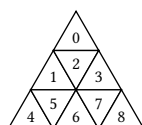
A **position** is a sticker's index on its face, as the face view draws it; `labels: true` writes the indices on the picture. $N \times N$ cubes also take `(row, col)` from the top left. Side faces are drawn with U at the top; U from above with B at the top; D from below with F at the top. On the pyraminx, D has the B vertex at the top. Megaminx faces always have a vertex at the top, so its side faces have U at the upper right. On the Square-1, U and D hold one sticker per piece, clockwise from the slice at the back, so their count changes with the state; the sides of the pieces and the equator are drawn but not indexed.



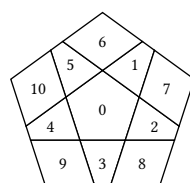
3x3, face U



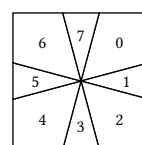
skewb, face U



pyraminx, face F



megaminx, face F



square1, face U

3 Drawing a state

```
draw(c, view: auto, mask: auto,
      sticker: 6mm, gap: 0pt, stroke: 0.5pt + black, radius: 0pt, body: none,
      palette: colors, hidden: auto,
      face: auto, top: auto, tip: auto,
      sides: auto, side-length: 0.35, labels: false,
      arrows: (), arrow-color: black, arrow-thickness: 1.6pt, arrow-head: 0.3,
      spacing: auto, options: (:))
```

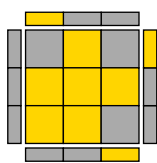
`view` picks the kind of picture and which stickers are hidden; `auto` is full, or layers on the Square-1. `mask` overrides the hiding. The rest is appearance and means the same in every view; a view ignores what it does not use, so one set of parameters serves all views.

3.1 Views

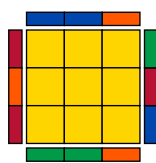
A view is a kind of picture plus a default **mask** (which stickers are greyed out). `face`, `net`, `full` and `layers` are the generic views; the others fix some arguments for a common diagram, so `view: "oll"` already hides everything but the top color. Asking a puzzle for a view it lacks is an error that lists the available ones.

view	picture	hidden by default	puzzles
"face"	one face straight on, chosen with face:	nothing	all
"net"	unfolded net	nothing	all
"full"	3D	nothing	$N \times N$ cubes, skewb, pyraminx, megaminx
"f2l"	full with a mask	every piece carrying the top color	$N \times N$ cubes

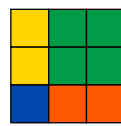
view	picture	hidden by default	puzzles
"oll"	ll with a mask	everything but the top color	N×N cubes
"ll"	face with face: "U", sides: true	nothing	N×N cubes
"tip"	3D, looking down the vertex chosen with tip:	nothing	pyraminx
"layers"	both layers straight on	nothing	Square-1
"obl"	layers with a mask, caps only	everything but the U and D colors	Square-1
"cs"	layers with a mask, caps only	everything (shape only)	Square-1



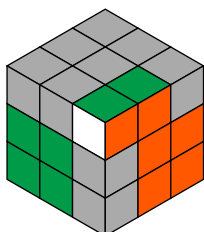
"oll"



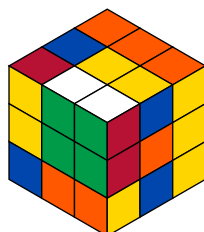
"ll"



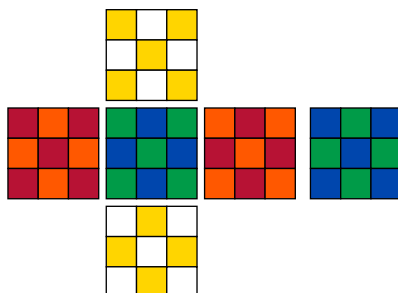
"face", face: "F"



"f2l"

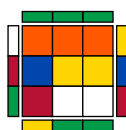


"full"

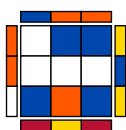


"net"

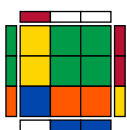
Straight on. One face head-on, oriented as in Section 2.3. sides: true adds every neighbouring sticker that touches the face as a strip outside the shared edge, so U with sides is the whole last layer seen from above. sides: auto is off for face, on for ll and oll.



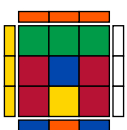
face: "U"



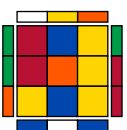
face: "D"



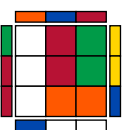
face: "F"



face: "B"

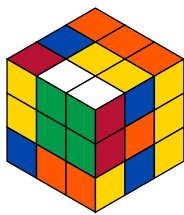


face: "R"

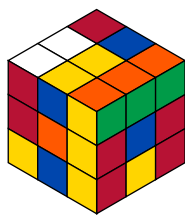


face: "L"

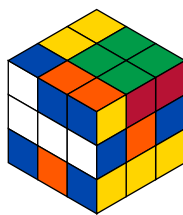
3D. An orthographic view with face in front and top on top, F and U by default. The camera keeps its place relative to these two, so they must sit like F and U: adjacent faces, or on the pyraminx a face and one of its vertices (top: auto picks the first that fits face). N×N cubes and the skewb: from the corner between face, top and the face to their right. Pyraminx: from in front of face, slightly below, the top vertex up; tip straight down the vertex named by tip:, the usual last-layer picture. Megaminx: straight down face (Section 4.3).



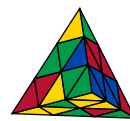
default: F, U



face: "R"



face: "D", top: "F"



pyraminx face: "D"

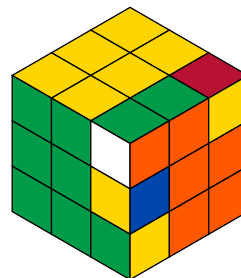
Net. Every face unfolded, as pictured in Section 2. The only view that shows every sticker.

Layers. The Square-1's two layers straight on: U from above, D from below, each with the sides of its pieces around it, the slice on the right marked by a grey line; side by side, or top to bottom with options: (direction: "vertical"). The equator only appears in net.

3.2 Mask

Leave mask: out for the view's default. mask: none shows every sticker; mask: f uses your own function from a state to a state. Hidden stickers are drawn in the hidden color.

```
#draw(case("U R U' R'"), view: "f2l", mask: none)
```



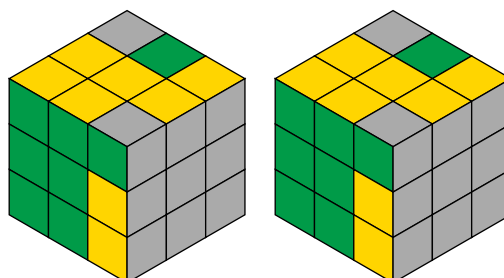
These functions return a state with some stickers hidden. They only ever hide, so they combine in any order, and work on every puzzle:

keep-colors(c, "yellow")	keep only these color name(s)
hide-faces(c, ("D", "B"))	hide whole faces
hide-pieces(c, containing: "yellow")	hide every piece that has a sticker of that color
mask(c, info => ..)	keep stickers for which the function is true; info has face, index, color, piece, plus row, col on N×N cubes

Pass one as mask: (as a function, so draw fills in the state) or apply it to the state first; the result is the same:

```
#draw(case("R U R' U'"), view: "full",
  mask: c => keep-colors(c, ("yellow",
    "green")))

#draw(keep-colors(case("R U R' U'"),
  ("yellow", "green")),
  view: "full")
```



3.3 Appearance parameters

Lengths must be absolute (mm, pt, cm). Sizes are in **units**: one unit is the edge of a cube sticker, or of the megaminx centre pentagon (at the default cut); a skewb face and a pyraminx edge are 3 units.

parameter	default	meaning	used by
sticker	6mm	length of one unit	all
gap	0pt	space between stickers; body shows through it	all
stroke	0.5pt + black	sticker outline: one Typst stroke (thickness + color), a thickness, a color, a dictionary, or none	all
radius	0pt	corner radius of square stickers	all but 3D
body	none	color behind the stickers, seen through gaps and the margin around a face	all
palette	colors	color name → color (Section 3.3.1)	all
hidden	auto	color of masked stickers; auto = the palette's hidden entry	all
face	auto	face view: the face shown, auto = U (F on the pyraminx); full: the face in front, auto = F; not on the Square-1	face, full
top	auto	full: the face on top (pyraminx: the vertex); auto = U, or the first that fits face	full
tip	auto	vertex the pyraminx tip view looks down: U, L, R or B; auto = U	tip
sides	auto	draw the neighbours' stickers as strips; auto = off for face, obl, cs, on for ll, oll, layers	face, ll, oll, layers, obl, cs
side-length	0.35	strip thickness in units	face, ll, oll, layers, obl, cs, net
labels	false	true writes each sticker's index on it; "faces" writes the face names instead	face, ll, oll, layers, obl, cs, net
arrows	()	arrows between positions (Section 3.3.3)	face, ll, oll
arrow-color	black	default arrow color	face, ll, oll
arrow-thickness	1.6pt	default arrow line thickness	face, ll, oll
arrow-head	0.3	default arrow head length in units	face, ll, oll

parameter	default	meaning	used by
spacing	auto	distance between faces; auto = a quarter unit (half a unit between layers)	net, layers, obl, cs
options	(:)	settings of one puzzle, listed below; other keys are errors	see below

Like `cube(options:)`, `draw(options:)` holds the settings that exist for one puzzle only:

puzzle	option	default	meaning
Square-1	slices	auto	mark the slice with a grey line; auto = on except for face
Square-1	direction	"horizontal"	layers side by side, or "vertical" for top to bottom
Square-1	turn	false	turn each layer so that its slice is vertical, as net always does

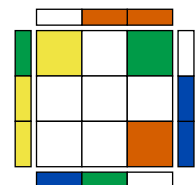
3.3.1 Palette

The palette maps color **names** to colors (the second lookup in Section 2.2). It belongs to the picture, not the state, so one state can be drawn with several palettes. The default colors has the six cube colors, five more for the megaminx, black, and hidden for masked stickers (grey is an ordinary color):

						
white	yellow	orange	red	green	blue	grey
						
purple	pink	lime	lightblue	beige	black	hidden

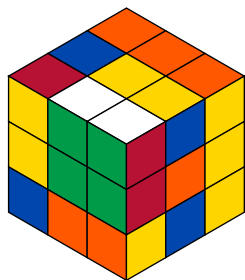
Spread colors to change some entries, for example for print:

```
#let print = (..colors,
  red: rgb("#d55e00"), orange: rgb("#f0e442"),
  yellow: rgb("#ffffff"), white: rgb("#999999"))
#draw(case(sune), view: "ll", palette: print)
```

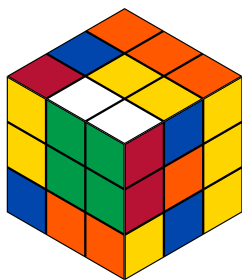


3.3.2 Outlines and gaps

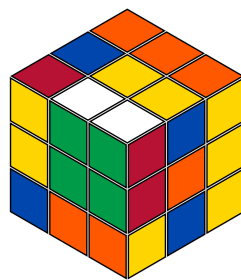
stroke is the sticker outline, one Typst stroke value (0.5pt + black is a single argument, as in rect). gap shrinks each sticker towards its centre by half the gap; the space shows body, which also fills the margin between a face and its strips. So stickers are separated by outlines, by gaps, or both:



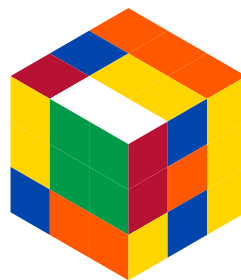
default
stroke: 0.5pt + black
gap: 0pt



gaps only
stroke: none
gap: 1pt, body: black

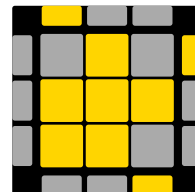


both
gap: 1pt
body: luma(230)



neither
stroke: none
gap: 0pt

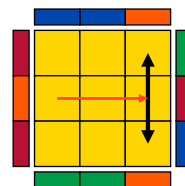
```
#draw(case(sune), view: "oll",
  stroke: none, gap: 1pt, body: black,
  radius: 1pt, side-length: 0.5)
```



3.3.3 Arrows

Arrows join positions (Section 2.3) on the shown face of any puzzle. Each is a pair (from, to) or a dictionary with from, to and optionally double: true, color, thickness and head, the last three overriding the arrow-* defaults for that arrow:

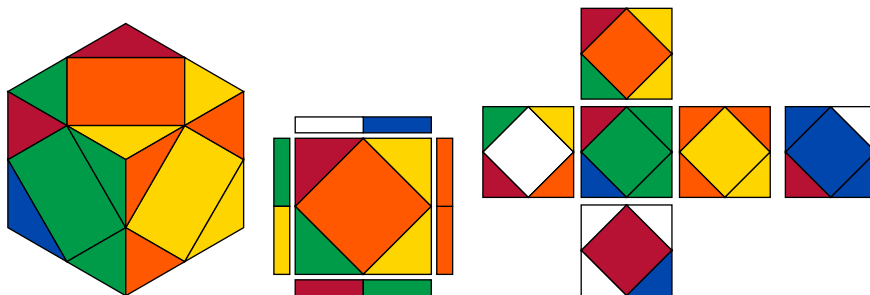
```
#draw(case(tperm), view: "ll", arrows: (
  (from: (0, 2), to: (2, 2), double: true),
  (from: (1, 0), to: (1, 2),
    color: red, thickness: 1pt, head: 0.2),
  ))
```



4 Other puzzles

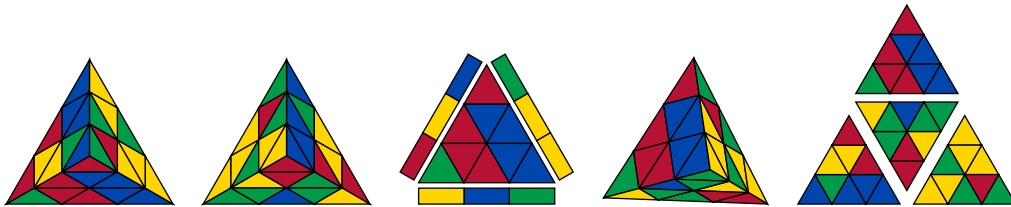
4.1 Skewb

```
#let c = cube(event: "skewb",
  scramble: "R L U B' R' L")
#draw(c, view: "full")
#draw(c, view: "face", face: "U", sides: true)
#draw(c, view: "net", sticker: 4mm)
```



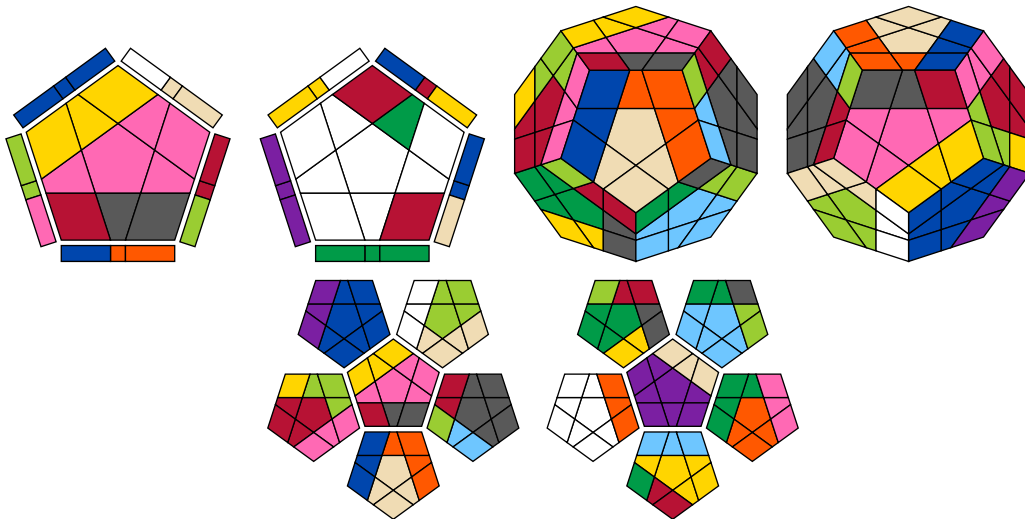
4.2 Pyraminx

```
#let c = cube(event: "pyraminx",
  scramble: "U L R' B u l'")
#draw(c, view: "tip")
#draw(c, view: "tip", tip: "B")
#draw(c, view: "face", sides: true)
#draw(c, view: "full")
#draw(c, view: "net", sticker: 4.5mm)
```



4.3 Megaminx

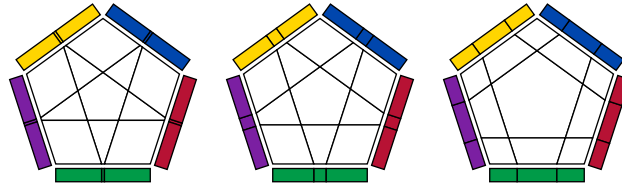
```
#let c = cube(event: "megaminx",
  scramble: "R++ D-- R++ D++ U F' BL2")
#draw(c, view: "face", sides: true)
#draw(case("R U R' U'", event: "megaminx"),
  view: "face", sides: true)
#draw(c, view: "full", sticker: 4mm)
#draw(c, view: "full", face: "U", sticker: 4mm)
#draw(c, view: "net", sticker: 3mm)
```



full looks straight down face with top up and shows it with its five neighbours: F by default, or face: "U" for the top half (then F is on top).

cut is the centre pentagon's inradius as a fraction of the face's: 0.4 by default, between 0.05 and 0.95. Keep it above 0.31, below which the edge stickers no longer reach the edge of the face. It changes the sticker shapes and the layer depth, so it is a cube() option, not a draw() one:

```
#for cut in (0.33, 0.4, 0.6) {
  draw(cube(event: "megaminx", options: (cut: cut)),
    view: "face", sides: true, sticker: 5mm)
}
```

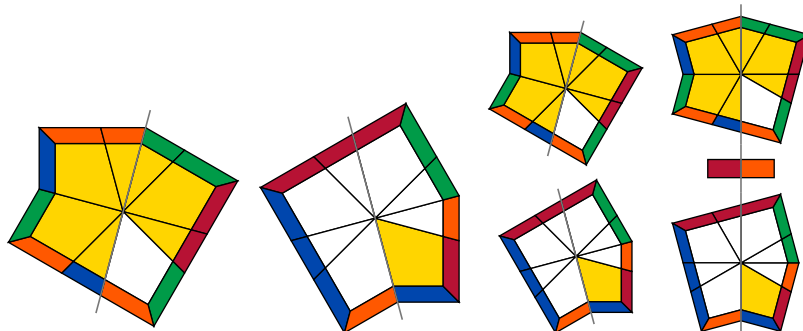


4.4 Square-1

Held the WCA way: yellow on top, white below, red in front, the slice through F and B with the smaller front part on the left. The shape follows the state.

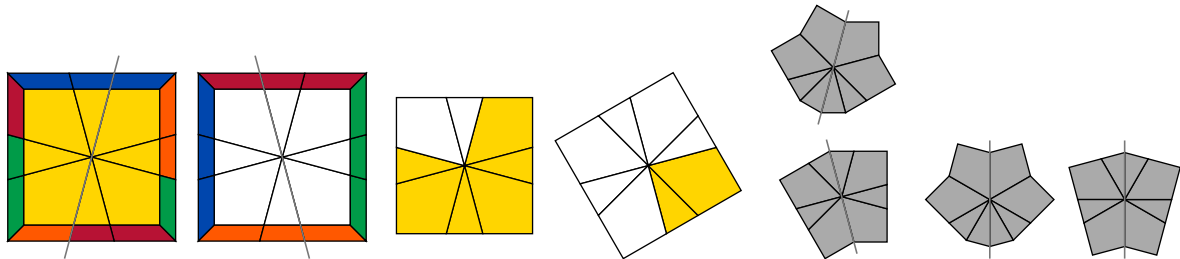
Views (Section 3.1). layers: U from above and D from below, each with the sides of its pieces, the slice on the right marked by a grey line. net: the layers turned so that the slice is vertical, with the front of the equator between them: two reds of different lengths when it is in place, a red and an orange of the same length when it is flipped. obl keeps only the U and D colors (the orientation diagram); cs hides every sticker (the cube-shape diagram); both show only the caps unless sides: true. face: and top: do not apply.

```
#let c = cube(event: "square1",
  scramble: "(0,-1)/ (3,0)/ (4,0)/ (6,0)/ (3,0)/")
#draw(c, view: "layers")
#draw(c, view: "layers", sticker: 4mm,
  options: (direction: "vertical"))
#draw(c, view: "net", sticker: 4mm)
```



Options (Section 3.3). direction: "vertical" stacks the layers, slices: false removes the line, turn: true turns each layer so that the line is vertical:

```
#let adj = "/" (3,-3) / (3,0) / (-3,0) / (0,3) / (-3,0) /"
#let sk = cube(event: "square1", scramble: "/" (3,0) / (1,2) /") // scallop / kite
#draw(case(adj, event: "square1"), view: "layers")
#draw(cube(event: "square1", scramble: "(0,-1) / (-3,0) /"),
  view: "obl", options: (slices: false))
#draw(sk, view: "cs", sticker: 4mm, options: (direction: "vertical"))
#draw(sk, view: "cs", sticker: 4mm, options: (turn: true))
```

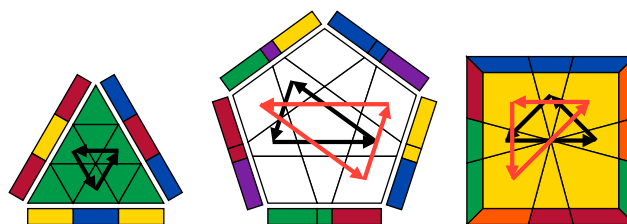


Moves. A / with a corner across the slice is a compile error naming the moves done so far. is-solved means cube shape with every piece in place, up to turning the layers.

4.5 Arrows on other puzzles

Arrows (Section 3.3.3) take the positions of Section 2.3. Below: an edge cycle on the pyraminx, the T perm on the megaminx, and the Square-1 algorithm above, which cycles three corner-edge blocks; corner cycles in red:

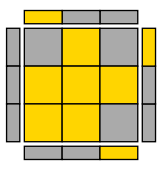
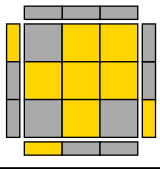
```
#let corner(a, b) = (from: a, to: b, color: red)
#draw(case("L' U L U R U R'", event: "pyraminx"),
  view: "face", sides: true, arrows: ((3, 1), (1, 6), (6, 3)))
#draw(case(tperm, event: "megaminx"), view: "face", sides: true,
  arrows: ((4, 2), (2, 5), (5, 4),
    corner(8, 7), corner(7, 10), corner(10, 8)))
#draw(case(adj, event: "square1"), view: "face", sides: true,
  arrows: ((5, 1), (1, 7), (7, 5),
    corner(4, 0), corner(0, 6), corner(6, 4)))
```



5 Recipes

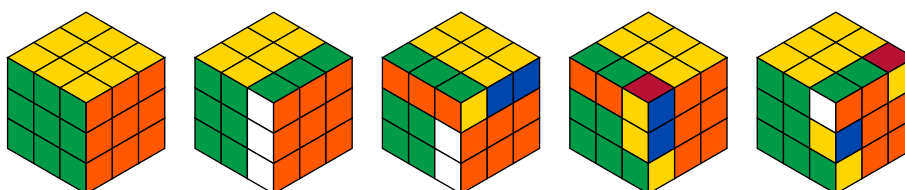
5.1 Algorithm sheet

```
#let algs = (
  ("Sune", "R U R' U R U2 R'"),
  ("Anti-Sune", "R U2 R' U' R U'
R'"),
)
#table(
  columns: 3, align: horizon,
  ..algs.map(((name, alg)) => (
    name, draw(case(alg), view:
"oll", sticker: 5mm), raw(alg)
  )).flatten(),
)
```

Sune		R U R' U R U2 R'
Anti-Sune		R U2 R' U' R U' R'

5.2 Step by step

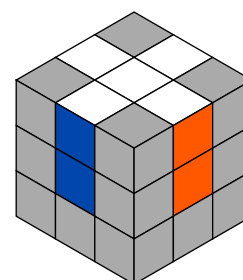
```
#let steps = ("R", "U", "R'", "U'")
#let states = range(steps.len() + 1)
  .map(k => cube(scramble: steps.slice(0, k).join(" ")))
#grid(columns: 5, gutter: 4mm,
  ..states.map(c => draw(c, view: "full", sticker: 4mm)))
```



5.3 Cross

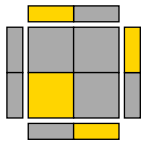
Only the white edges and the centres. hide-pieces hides exactly the pieces carrying white, so a sticker it turned into none is on one; a row or col of 1 rules out corners. x2 puts white on top, where full can see it.

```
#let c = apply(cube(), "x2")
#let rest = hide-pieces(c, containing: "white")
#let cross = mask(c, info =>
  (info.row == 1 and info.col == 1) // every centre ..
  or (rest.faces.at(info.face).at(info.index) == none
    and (info.row == 1 or info.col == 1))) // .. white edges
#draw(cross, view: "full")
```

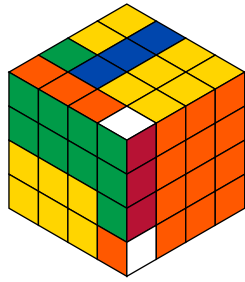


5.4 Other cube sizes

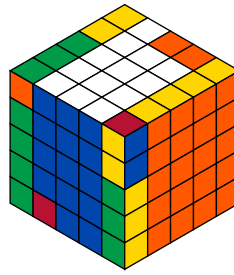
Every cube view works for any "NxN". Wide moves take a layer count (3Rw); M, E, S turn all middle layers.



"2x2", "oll"



"4x4", "Rw U 3Rw' F2"



"5x5", "3Rw U 3Rw' M2"

6 Reference

function

`cube(event:, scramble:, inverted:, scheme:, options:)`
`case(alg, event:, scheme:, options:)`
`solved(event:, scheme:, options:)`
`apply(c, alg)`
`is-puzzle(x)`
`parse(alg, event:, options:)`

`inverse(alg, event:, options:)`
`to-string(moves, event:, options:)`
`face(c, name), sticker(c, name, ..pos), is-solved(c)`
`keep-colors, hide-faces, hide-pieces, mask`
`draw(c, view:, mask:, ..)`
`draw-face, draw-3d, draw-net, draw-layers`

`views`

`colors, default-scheme`
`puzzles`

purpose

build a state

 the state an algorithm solves
 a solved state
 apply moves, returns a new state
 whether a value is a state
 algorithm string → move array (apply accepts both)
 the inverse, as a move array
 move array → notation
 read a state (Section 2.3)

 hide stickers (Section 3.2)
 draw a state (Section 3)
 the renderers behind draw; draw-3d also takes a camera: (dir:, up:)
 the view table: kind, fixed arguments and default mask of each view
 the default palette and cube scheme
 the puzzle definitions, keyed cube, skewb, pyraminx, megaminx, square1